

T.Y.B.Sc. (Computer Science) – Sem V

CS-3010-MJ-P: Database Technologies

Complete Practical Examination Slip Solutions – 2026-27

Based on the uploaded Savitribai Phule Pune University practical examination paper. The paper contains 25 slips, each with Q1 (10M), Q2 (20M) and Viva (5M).

Slip 1

Q1. JSON – Student Record

Solution

```
{
  "name": "Amit Patil",
  "rollNo": 101,
  "age": 21,
  "department": "Computer Science",
  "subjects": ["DBMS", "Java", "Operating System"],
  "address": {"city": "Nashik", "pin": 422001}
}
```

Q2. Department document database

Collections: students, courses, marks.

Example student document:

```
{ "_id": 1, "firstName": "Amit", "lastName": "Patil", "age": 21, "percentage": 72 }
```

Create 10+ documents in each collection. A student may register for multiple courses, so marks can contain studentId and courseId.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
use Department
db.students.countDocuments()

db.students.find({}, {name:1, age:1}).sort({age:-1}).limit(5)

db.students.deleteOne({firstName:"S", lastName:"L"})

db.students.find({percentage:{>50}})

db.dropDatabase()
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 2

Q1. BSON vs JSON

Solution

BSON (Binary JSON) is a binary-encoded representation of JSON-like documents. JSON is a text-based data interchange format.

JSON example:

```
{"name":"Amit", "age":21}
```

Additional BSON data types include:

1. ObjectId: ObjectId("507f1f77bcf86cd799439011")
2. Date: ISODate("2026-10-07T00:00:00Z")
3. Decimal128: Decimal128("12345.67")
4. Binary: BinData(0,"SGVsbG8=")
5. Int32 / Int64: NumberInt(25), NumberLong(123456789)

BSON is commonly used internally by MongoDB and supports types that standard JSON does not directly provide.

Q2. Hospital document database

Collections: hospitals, doctors, people/reviews. Include specialization arrays and doctor visiting schedules.

Example:

```
{hospital:"Jehangir Hospital", specializations:["Pediatric","Gynaec","Orthopaedic"]}
```

Queries:

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.hospitals.find({specializations:"Pediatric"},{name:1})

db.doctors.find(
  {hospital:"Jehangir Hospital", "visitingDays":"Monday"},
  {name:1}
)

db.hospitals.find({
  specializations:{$size:{$gte:2}},
  doctors:"Dr. Kale"
},{name:1})

db.reviews.find({
  hospital:"Jehangir Hospital",
  rating:{$gte:3}
},{personName:1})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 3

Q1. Doctor information – Graph model

Solution

```
Nodes:
(:Doctor {name, id})
(:Hospital {name, city})
(:Department {name})
(:Patient {name, id})

Relationships:
(Doctor)-[:WORKS_IN]->(Hospital)
(Doctor)-[:SPECIALIZED_IN]->(Department)
(Patient)-[:CONSULTS]->(Doctor)
(Patient)-[:ADMITTED_IN]->(Hospital)
```

```
CREATE (d:Doctor {name:"Dr. Kale"})
CREATE (h:Hospital {name:"Jehangir Hospital"})
CREATE (dep:Department {name:"Cardiology"})
CREATE (p:Patient {name:"Amit"})
```

```
CREATE (d)-[:WORKS_IN]->(h)
CREATE (d)-[:SPECIALIZED_IN]->(dep)
CREATE (p)-[:CONSULTS]->(d)
CREATE (p)-[:ADMITTED_IN]->(h)
```

```
MATCH (p:Patient) RETURN p
MATCH (d:Doctor)-[:WORKS_IN]->(h:Hospital) RETURN d,h
MATCH (d:Doctor) RETURN d
```

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

Q2. Department document database – same model and queries as Slip 1.

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 4

Q1. JSON – Student Record

Solution

```
{
  "name": "Amit Patil",
  "rollNo": 101,
  "age": 21,
  "department": "Computer Science",
  "subjects": ["DBMS", "Java", "Operating System"],
  "address": {"city": "Nashik", "pin": 422001}
}
```

Q2. Online Mobile Shopping

Collections: mobiles, customers, orders. Insert 10+ documents in each.

Example mobile:

```
{brand:"Samsung", model:"J6", ram:3, rom:32, price:12000}
```

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.mobiles.find({ram:3, rom:32})
db.orders.find({mobile:"Samsung J6"},{customerName:1})
db.orders.find({mobile:"iPhone 7plus"},{customerName:1}).sort({customerName:1})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 5

Q1. Hotel – Graph model

Solution

Nodes: Customer, Room, Hotel, City.

Relationships:

(Customer)-[:BOOKS]->(Room)

(Room)-[:BELONGS_TO]->(Hotel)

(Hotel)-[:LOCATED_IN]->(City)

```
CREATE (c:Customer {name:"Amit"})
CREATE (r:Room {number:101})
CREATE (h:Hotel {name:"Hotel ABC"})
CREATE (city:City {name:"Pune"})
CREATE (c)-[:BOOKS]->(r)
CREATE (r)-[:BELONGS_TO]->(h)
CREATE (h)-[:LOCATED_IN]->(city)
```

```
MATCH (h:Hotel) RETURN h
```

```
MATCH (h:Hotel)-[:LOCATED_IN]->(:City {name:"Pune"}) RETURN h
```

```
MATCH (c:Customer) RETURN c
```

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

Q2. Inventory document database

Collections: items, categories, warehouses. Include tags array, status, quantity and height.

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 6

Q1. JSON – Bank account

Solution

```
{
  "holder": {"name": "Amit Patil", "age": 21},
  "account": {"accountNo": "AC1001", "type": "Savings", "balance": 50000},
  "transactions": [
    {"date": "2026-10-01", "type": "Credit", "amount": 10000},
    {"date": "2026-10-03", "type": "Debit", "amount": 2500}
  ]
}
```

Q2. Furniture Shopping

Collections: furniture, customers, purchases. Insert 10+ documents each.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.furniture.find({type:"Sofa"})
db.furniture.find({price:{$lt:20000}})
db.furniture.find({company:"IKEA"})
db.customers.find({name:"Neha"})
db.purchases.find({rating:5})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 7

Q1. Society – Graph model

Solution

```
Nodes: Person, City, Project.  
Relationships:  
(Person)-[:FRIEND_OF]->(Person)  
(Person)-[:SIBLING_OF]->(Person)  
(Person)-[:PARENT_OF]->(Person)  
(Person)-[:STAYS_IN]->(City)  
(Person)-[:WORKS_ON]->(Project)
```

```
MATCH (p:Person)-[:PARENT_OF]->>() RETURN DISTINCT p.name
```

```
MATCH (p:Person)-[:WORKS_ON]->(pr:Project {name:"Finance"}) RETURN p.name
```

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

Q2. Laptop Shopping
Collections: laptops, customers, purchases. Insert 10+ documents each.

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 8

Q1. JSON - Employee

Solution

```
{
  "name": "Rahul Sharma",
  "employeeId": 1001,
  "job": { "title": "Developer", "department": "IT", "salary": 60000 },
  "projects": [
    { "name": "ERP", "role": "Developer" },
    { "name": "Mobile App", "role": "Team Member" }
  ]
}
```

Q2. Inventory - same model as Slip 5.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.items.find({status:"D", quantity:{$gt:30}})
db.items.find({$expr:{$eq:[{$size:"$tags"},3]}})
db.items.find({$or:[{status:"A"},{quantity:{$lt:30},height:{$gt:10}}]})
```

Q3. Viva - Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 9

Q1. JSON – Mobile

Solution

```
{
  "brand": "Samsung",
  "model": "Galaxy M14",
  "specifications": {"RAM": "6GB", "ROM": "128GB", "processor": "Exynos"},
  "price": 14999,
  "ratings": [5, 4, 5, 4]
}
```

Q2. Online Vehicle Shopping

Collections: vehicles, customers, purchases.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.vehicles.find({type:"SUV"})
db.vehicles.find({price:{$lt:1000000}})
db.vehicles.find({company:"Tata"})
db.customers.find({name:"Amit"})
db.purchases.find({rating:5})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 10

Q1. College – Graph model

Solution

Nodes: Student, Course, Teacher, Department, College.

Relationships:

(Student)-[:ENROLLED_IN]->(Course)

(Teacher)-[:TEACHES]->(Course)

(Teacher)-[:WORKS_IN]->(Department)

(Course)-[:OFFERED_BY]->(Department)

(Department)-[:BELONGS_TO]->(College)

MATCH (c:Course) RETURN c

MATCH (s:Student {name:"Amit"})-[:ENROLLED_IN]->(c:Course) RETURN c

MATCH (s:Student) RETURN s

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

Q2. Tours document database

Collections: tours/packages, industries, customers/expenses. Insert 10+ documents.

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 11

Q1. Clothing Brand – Graph model

Solution

```
Nodes: Mall, Section, Apparel, Brand, Design, Size, Staff.  
Relationships:  
(Mall)-[:HAS_SECTION]->(Section)  
(Section)-[:CONTAINS]->(Apparel)  
(Apparel)-[:FROM_BRAND]->(Brand)  
(Apparel)-[:HAS_DESIGN]->(Design)  
(Apparel)-[:AVAILABLE_IN]->(Size)  
(Staff)-[:ASSIGNED_TO]->(Section)
```

MATCH (s:Section {name:"Female"})-[:CONTAINS]->(a:Apparel) RETURN DISTINCT a.type

MATCH (st:Staff)-[:ASSIGNED_TO]->(s:Section {name:"Kids"}) RETURN st.name

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

Q2. Watch Shopping
Collections: watches, customers, purchases.

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 12

Q1. JSON - Company

Solution

```
{
  "companyName": "ABC Technologies",
  "companyId": "C101",
  "address": {"city": "Pune", "pin": 411001},
  "employees": [
    {"id": 1, "name": "Amit", "role": "Developer"},
    {"id": 2, "name": "Sneha", "role": "Tester"}
  ]
}
```

Q2. Online Bike Shopping

Collections: bikes, customers, purchases.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.bikes.find({type:"Sports"})
db.bikes.find({price:{$lt:200000}})
db.bikes.find({company:"Honda"})
db.customers.find({name:"Sneha"})
db.purchases.find({rating:5})
```

Q3. Viva - Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 13

Q1. JSON – Online shopping order

Solution

```
{
  "orderId": "01001",
  "customer": { "name": "Amit", "email": "amit@example.com" },
  "payment": { "mode": "UPI", "amount": 2500, "status": "Paid" },
  "products": [
    { "productId": "P1", "name": "Mouse", "qty": 1, "price": 800 },
    { "productId": "P2", "name": "Keyboard", "qty": 1, "price": 1700 }
  ]
}
```

Q2. Online Book Store

Collections: books, authors, publishers, customers, purchases.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.books.find({author:{ $in: ["R. K. Narayan", "Chetan Bhagat"]}})
db.books.find({pages:{ $gt: 300}})
db.books.find({price:{ $gte: 300, $lte: 700}})
db.purchases.aggregate([
  { $group: { _id: "$customer", bookCount: { $sum: 1 } } },
  { $match: { bookCount: { $gt: 1 } } }
])
db.books.find().sort({rating: -1})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 14

Q1. JSON - Vehicle

Solution

```
{
  "vehicle": {"id": "V101", "model": "Nexon", "company": "Tata"},
  "specifications": {"engine": "1199cc", "fuel": "Petrol"},
  "features": ["ABS", "Airbags", "Touchscreen"],
  "companyInfo": {"city": "Mumbai"}
}
```

Q2. Online Camera Shopping

Collections: cameras, customers, purchases.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.cameras.find({resolution:24})
db.cameras.find({price:{$gt:40000}})
db.cameras.find({type:"Mirrorless"})
db.purchases.find({company:"Canon"}, {customerName:1})
db.cameras.find({rating:{$gt:4}})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 15

Q1. University – Graph model

Solution

Nodes: University, Department, Course, Person.
Relationships:
(Department)-[:OFFERS]->(Course)
(Course)-[:RECOMMENDED_BY]->(Person)
(Department)-[:PART_OF]->(University)

MATCH (d:Department {name:"Chemistry"})-[:OFFERS]->(c:Course) RETURN c.name

MATCH (d:Department) RETURN d

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

Q2. Online Shoe Shopping
Collections: shoes, customers, purchases.

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 16

Q1. Restaurant – Graph model

Solution

Nodes: Customer, Food, Restaurant, City.
Relationships:
(Customer)-[:ORDERS]->(Food)
(Food)-[:SERVED_AT]->(Restaurant)
(Restaurant)-[:LOCATED_IN]->(City)

MATCH (c:Customer {name:"Priya"})-[:ORDERS]->(f:Food) RETURN f

MATCH (f:Food)-[:SERVED_AT]->(r:Restaurant {name:"Domino's"}) RETURN f

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

Q2. Online Camera Shopping – same collections as Slip 14.

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 17

Q1. JSON array – Product catalog

Solution

```
[
  {"product_id":"P01","product_name":"Mouse","category":"Computer","price":800,"stock_quantity":50},
  {"product_id":"P02","product_name":"Keyboard","category":"Computer","price":1500,"stock_quantity":30},
  {"product_id":"P03","product_name":"Monitor","category":"Display","price":12000,"stock_quantity":10}
]
```

Q2. Transaction document database

Collections: users, items, transactions. Each transaction stores user, items, paymentType and totalAmount.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.transactions.find({user:"John"})
db.transactions.find({paymentType:"Debit Card"})
db.transactions.aggregate([
  {$group:{_id:"$paymentType", totalPayment:{$sum:"$totalAmount"}}}
])
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 18

Q1. JSON array – Product catalog

Solution

```
[
  {"product_id":"P01","product_name":"Mouse","category":"Computer","price":800,"stock_quantity":50},
  {"product_id":"P02","product_name":"Keyboard","category":"Computer","price":1500,"stock_quantity":30},
  {"product_id":"P03","product_name":"Monitor","category":"Display","price":12000,"stock_quantity":10}
]
```

Q2. Sales system

Collections: products, customers, orders, invoices. An invoice is generated when an order is processed.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.products.find({})

db.orders.find({value:{$gt:10000}})

// Example join-style lookup of invoice with orders:
db.orders.aggregate([
  {$match:{value:{$gt:10000}}},
  {$lookup:{
    from:"invoices",
    localField:"orderId",
    foreignField:"orderId",
    as:"invoice"
  }}
])
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 19

Q1. Vehicle – Graph model

Solution

```
Nodes: VehicleType, Characteristic, Customer, Person.  
Relationships:  
(VehicleType)-[:HAS_CHARACTERISTIC]->(Characteristic)  
(Customer)-[:BUYS]->(VehicleType)  
(Person)-[:RECOMMENDS]->(VehicleType)  
(Person)-[:RATES]->(VehicleType)
```

```
MATCH (v:VehicleType {name:"Heavy  
Vehicle"})-[:HAS_CHARACTERISTIC]->(c:Characteristic)  
RETURN c
```

```
MATCH (c:Customer)-[:BUYS]->(v:VehicleType {name:"Light Vehicle"})  
RETURN c
```

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

Q2. Inventory – same model as Slip 5.

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 20

Q1. JSON – Student record

Solution

```
{
  "name": "Amit Patil",
  "rollNo": 101,
  "course": "B.Sc. Computer Science",
  "marks": [
    {"subject": "DBMS", "marks": 78},
    {"subject": "Java", "marks": 82},
    {"subject": "OS", "marks": 75}
  ]
}
```

Q2. Online Electronics Shopping

Collections: Electronics, Customers, orders. Insert 10+ documents each.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.Electronics.find({brand:"Samsung"})
db.Customers.find().sort({name:1})
db.Electronics.find({brand:"Sony"})
db.Electronics.find({}, {productName:1, price:1, _id:0})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 21

Q1. JSON – Product

Solution

```
{
  "productName": "Laptop",
  "productId": "P101",
  "price": 60000,
  "category": "Electronics",
  "colors": ["Black", "Silver", "Grey"],
  "sellerAddress": {"city": "Nashik", "pin": 422001}
}
```

Q2. Newspaper database

Collections: newspapers, publishers, cities. Insert 10+ documents each.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.newspapers.find({city: "NASHIK"}, {name: 1})
db.newspapers.find({language: "Marathi"}, {name: 1})
db.publishers.countDocuments({state: "Gujrat"})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 22

Q1. JSON – Movie

Solution

```
{
  "movieName": "Example Movie",
  "releaseYear": 2026,
  "director": "Raj Patil",
  "actors": ["Amit", "Sneha", "Rahul", "Priya", "Neha", "Ravi", "Kiran", "Pooja", "Vikas", "Anita"],
  "production": {"company": "ABC Films", "city": "Mumbai"}
}
```

Q2. Company projects

Collections: employees, projects, assignments. Insert 10+ documents each.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.projects.find({project_type:"Software"},{name:1})
db.projects.find({duration:{>3}}, {name:1})
db.assignments.aggregate([
  {$match:{project:"AI"}},
  {$group:{_id:null,count:{$sum:1}}}
])
db.assignments.find({employee:"Mr. Patil"},{project:1})
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 23

Q1. JSON – Movie

Solution

```
{
  "movieTitle":"Example Movie",
  "movieId":"M101",
  "releaseYear":2026,
  "director":"Raj Patil",
  "actors":["Amit","Sneha","Rahul"],
  "productionCompany":{"name":"ABC Films","city":"Mumbai"}
}
```

Q2. Customer Loan database

Collections: customers, loans. A customer can take many loan types. Insert 10+ documents each.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.customers.find({name:/^D/},{name:1})

db.loans.aggregate([
  {$match:{city:"Pimpri"}},
  {$sort:{customerName:-1}},
  {$project:{customerName:1,_id:0}}
])

db.loans.find().sort({loanAmount:-1}).limit(1)
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 24

Q1. JSON – Student

Solution

```
{
  "name": "Amit Patil",
  "rollNo": 101,
  "dateOfBirth": "2005-05-12",
  "gender": "Male",
  "subjects": ["DBMS", "Java", "OS"],
  "contact": {"email": "amit@example.com", "mobile": "9876543210"}
}
```

Q2. Sales system

Collections: products, customers, orders, invoices. Insert 10+ documents each.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
db.products.find({})

db.orders.find({value:{$gt:20000}})

db.orders.aggregate([
  {$match:{customerName:"Mr. Rajiv"}},
  {$lookup:{
    from:"invoices",
    localField:"orderId",
    foreignField:"orderId",
    as:"invoice"
  }}
])
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.

Slip 25

Q1. JSON – Library member

Solution

```
{
  "memberName": "Amit Patil",
  "memberId": "L101",
  "age": 21,
  "borrowedBooks": ["Book A", "Book B"],
  "membershipType": "Premium",
  "address": {"city": "Nashik", "pin": 422001}
}
```

Q2. Student Competition database

Collections: students, competitions, participation/results. A student can participate in many competitions. Insert 10+ documents in each collection.

The examples use MongoDB syntax. For the examination, create the required collections first and insert at least 10 documents in each collection as instructed.

```
// i. Average number of students participating in each competition
db.participation.aggregate([
  {$group: {_id: "$competitionId", students: {$addToSet: "$studentId"}}},
  {$project: {
    competitionId: "$_id",
    averageStudents: {$size: "$students"}
  }}
])
```

```
// ii. Number of students in Programming competition
db.participation.aggregate([
  {$lookup: {
    from: "competitions",
    localField: "competitionId",
    foreignField: "_id",
    as: "competition"
  }},
  {$unwind: "$competition"},
  {$match: {"competition.name": "Programming"}},
  {$group: {_id: "$studentId"}},
  {$count: "numberOfStudents"}
])
```

```
// iii. First three winners of each competition
db.results.aggregate([
  {$sort: {competitionId: 1, rank: 1}},
  {$group: {
    _id: "$competitionId",
    winners: {$push: {studentId: "$studentId", rank: "$rank"}}
  }},
  {$project: {winners: {$slice: ["$winners", 3]}}}
])
```

Q3. Viva – Suggested questions and short answers

1. What is MongoDB? — A document-oriented NoSQL database that stores data as BSON documents.
2. What is a collection? — A group of MongoDB documents, similar to a table in a relational database.
3. What is a document? — A BSON record containing field-value pairs.
4. What is Cypher? — A declarative query language used with Neo4j graph databases.
5. What is JSON? — A text-based format for representing structured data.